

Department of Computer Engineering

Java Project Report

“Smart Inventory and Billing System”

SUBMITTED TO THE DEPARTMENT OF COMPUTER ENGINEERING

AISSMS IOIT

S. Y. BTech Engineering

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2025-26

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CERTIFICATE

This is to certify that the project report
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INDEX

Sr. No.	Chapter No	Title Of Chapter	Page no.
1	1	ABSTRACT	4
	1.1	Context	4
	1.2	Problem statement	4
	1.3	Objective	5
2	2	INTRODUCTION	6
	2.1	Project Introduction	6
	2.2	Importance of databases	7
	2.3	Scope of the project	8
3	3	SYSTEM DESIGN	9
	3.1	Design Objective	9
	3.2	System Architecture	9
	3.3	Flowchart	9
	3.4	Algorithm	11
	3.5	Database Design	12
	3.6	User Interface Design	12
4	4	IMPLEMENTATION	13
	4.1	Programming Languages and Tools Used	13
	4.2	Screenshots of the application	13
	4.3	Description of Database tables	20
5	5	TESTING	21
6	6	ADVANTAGES AND DISADVANTAGES	23
	6.1	Advantages	23
	6.2	Disadvantages	23
7	7	APPLICATIONS	24

8	8	CONCLUSION	25
9	9	REFERENCES	26

CHAPTER 1 – ABSTRACT

1.1 Context

In today's fast-paced financial world, managing personal investments has become increasingly complex. With a wide variety of investment options available — such as stocks, mutual funds, fixed deposits, real estate, and gold — individuals often find it difficult to track, plan, and optimize their portfolios. Most people either rely on manual spreadsheets or expensive financial advisors to manage their investments. A software-based Investment Planner system provides an accessible, automated, and efficient solution for individuals to manage and monitor their financial investments effectively.

1.2 Problem Statement

Many individuals lack the tools to plan their investments systematically. Manual tracking of investments leads to errors, missed opportunities, and difficulty in analyzing returns. There is no centralized platform where a user can input their investment details, view portfolio performance, set financial goals, and get a summary of returns across different asset types. This project aims to solve this problem by developing a Java-based Investment Planner that provides a unified interface for managing all personal investments.

1.3 Objective

- To develop a Java-based Investment Planner application with an intuitive GUI using Java Swing.
- To allow users to register and securely log in to their personal investment dashboard.
- To enable users to add, update, and delete various types of investments.
- To calculate and display returns (simple and compound interest) for each investment.
- To provide a summary view of the total portfolio value and asset-wise breakdown.
- To store all investment data persistently using a MySQL database via JDBC.
- To help users set financial goals and track progress toward them.

CHAPTER 2 – INTRODUCTION

2.1 Project Introduction

The **Investment Planner** is a desktop application developed using Core Java, Java Swing for the graphical user interface, and MySQL as the backend database connected via JDBC. The system is designed to help individual users manage their personal financial investments in a structured and efficient manner.

Users can register on the platform, log in securely, and access a personal dashboard where they can record different types of investments such as Fixed Deposits (FD), Recurring Deposits (RD), Mutual Funds, Stocks, and Gold. The system automatically calculates expected returns based on the investment type, principal amount, rate of interest, and tenure. It also provides a portfolio summary showing total invested amount, current value, and profit/loss.

Two user roles are supported:

- **Admin** – Manages user accounts, views all portfolios, and generates system-wide reports.
- **User/Investor** – Registers, adds investments, tracks returns, and sets financial goals.

2.2 Importance of Databases

A database is the backbone of the Investment Planner system. The importance of using MySQL in this project includes:

- **Persistent Storage** – All investment records, user data, and transaction histories are permanently stored.
- **Data Integrity** – Constraints and relations ensure accurate financial records.
- **Multi-user Support** – Multiple users can maintain their own separate investment portfolios.
- **Fast Retrieval** – SQL queries enable quick fetching of portfolio summaries and historical records.
- **Security** – User credentials and financial data are stored securely with hashed passwords.

2.3 Scope of the Project

The scope of this project includes:

- User registration, login, and profile management.
- Adding and managing investments across multiple categories (FD, RD, Stocks, Mutual Funds, Gold).
- Automatic calculation of simple and compound interest returns.
- Portfolio summary with total value and category-wise breakdown.
- Setting and tracking financial goals (e.g., saving ₹5,00,000 in 3 years).
- Admin panel for managing users and generating reports.
- The system is intended for individual/personal use and is not connected to live stock market APIs in its current version.

CHAPTER 3 – SYSTEM DESIGN

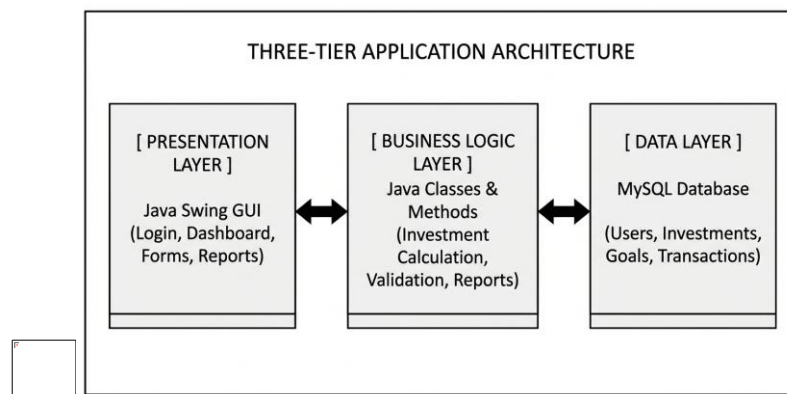
3.1 Design Objective

The design objective of the Investment Planner system is to create a modular, scalable, and user-friendly application. The design focuses on:

- Clear separation of the User Interface (Swing), Business Logic (Java classes), and Data Layer (MySQL/JDBC).
- Reusability of components such as investment calculators and report generators.
- A clean and intuitive GUI that requires minimal financial knowledge to operate.
- Secure handling of sensitive financial data.

3.2 System Architecture

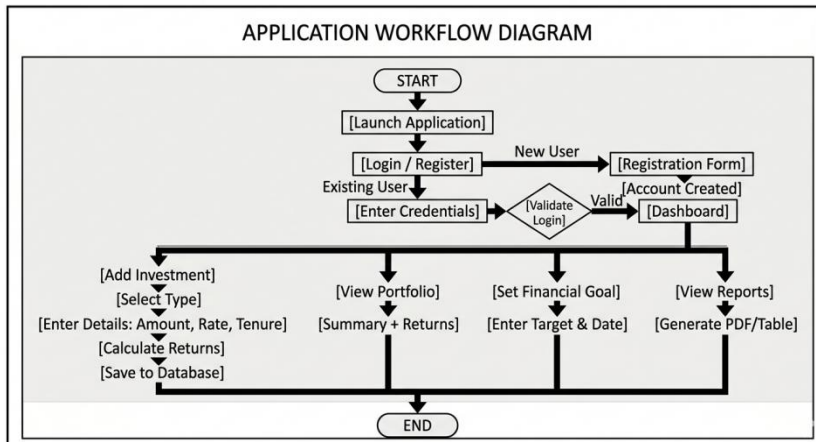
The system follows a **3-Tier Architecture**:



- **Presentation Layer** – Java Swing forms for login, registration, dashboard, add/view investments.
- **Business Logic Layer** – Java classes handle interest calculations, goal tracking, and validations.
- **Data Layer** – MySQL database stores all persistent data; JDBC is used for connectivity.

3.3 Flowchart

User Flow:



3.4 Algorithm

Algorithm for Investment Return Calculation:

1. START
2. Accept investment type from user (FD / RD / Stock / Mutual Fund / Gold)
3. Accept principal amount (P), rate of interest (R), tenure in years (T)
4. IF investment type is FD or simple interest based:

$$\text{Simple Interest (SI)} = (P \times R \times T) / 100$$

$$\text{Maturity Amount} = P + \text{SI}$$
5. IF investment type is Mutual Fund or Compound Interest based:

$$\text{Compound Interest (CI)} = P \times (1 + R/100)^T$$

$$\text{Maturity Amount} = \text{CI}$$
6. IF investment type is RD:

$$\text{Maturity Amount} = P \times n \times (1 + R/(4 \times 100))^{(4 \times T)}$$

where n = number of installments
7. Display Maturity Amount and Total Returns
8. Save investment record to database

9. Update portfolio summary

10. STOP

3.5 Database Design

Tables:

1. users:

Column	Type	Description
User_id	INT (PK, AI)	Unique user ID
name	VARCHAR(100)	Full name
email	VARCHAR(100)	Email (unique)
password	VARCHAR(255)	Hashed password
phone	VARCHAR(15)	Contact number
Created_at	DATETIME	Registration date

2. investments:

Column	Type	Description
investment_id	INT (PK, AI)	Unique investment ID
user_id	INT (FK)	References users table
Type	VARCHAR(50)	FD / RD / Stock / MF / Gold
principal	DECIMAL(12,2)	Invested amount
rate	DECIMAL(5,2)	Interest rate (%)
tenure	INT	Duration in years/months
maturity_amount	DECIMAL(12,2)	Calculated return
start_date	DATE	Investment start date

3. goals:

Column	Type	Description
goal_id	INT (PK, AI)	Unique goal ID

user_id	INT (FK)	References users table
goal_name	VARCHAR(100)	Name of financial goal
target_amount	DECIMAL(12,2)	Target savings amount
target_date	DATE	Date to achieve goal
achieved	BOOLEAN	Goal status

3.6 User Interface Design

The UI is built using **Java Swing** and consists of the following screens:

- **Login Screen** – Username/email and password fields with a login button.
- **Registration Screen** – New user sign-up form with validation.
- **Dashboard** – Overview panel showing total portfolio value, number of investments, and upcoming maturity dates.
- **Add Investment Form** – Dropdown for investment type, fields for amount, rate, tenure, and a Calculate button.
- **View Portfolio Screen** – Table listing all investments with returns and maturity dates.
- **Goal Tracker Screen** – Form to set financial goals with a progress bar showing completion.
- **Reports Screen** – Tabular and graphical summary of investments by category.
- **Admin Panel** – User management table and system-wide portfolio reports.

CHAPTER 4 – IMPLEMENTATION

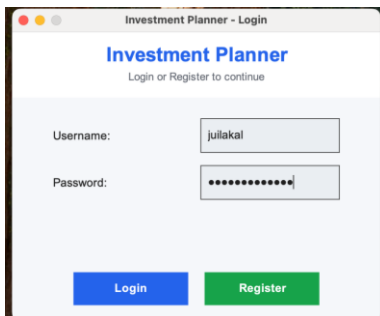
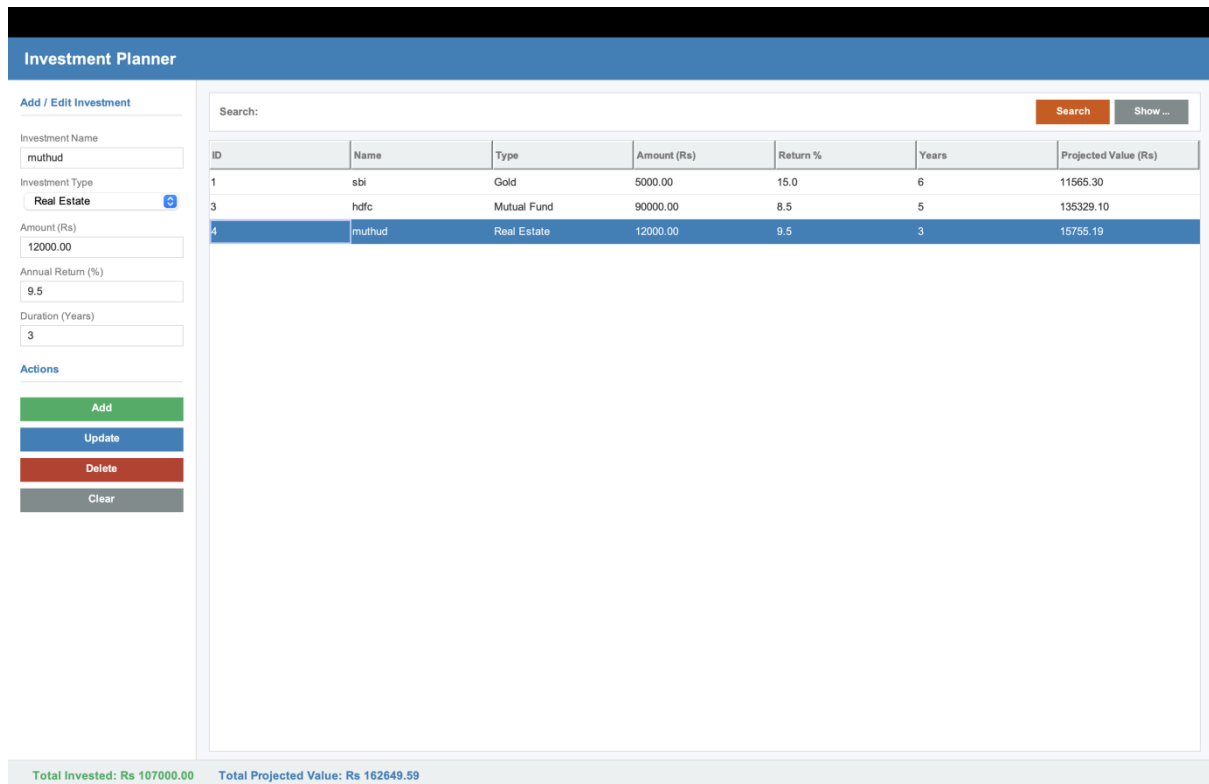
4.1 Programming Languages and Tools Used

Tool/Technology	Purpose
Java (JDK 8+)	Core programming language
Java Swing	GUI development
JDBC	Database connectivity
MySQL	Backend relational database
MySQL Workbench	Database design and management

Eclipse / IntelliJ IDEA IDE for development

JFreeChart (optional) Pie/bar charts for portfolio view

4.2 Screenshots of the Application

ID	Name	Type	Amount (Rs)	Return %	Years	Projected Value (Rs)
1	sbi	Gold	5000.00	15.0	6	11565.30
3	hdfc	Mutual Fund	90000.00	8.5	5	135329.10
4	muthud	Real Estate	12000.00	9.5	3	15755.19

Total Invested: Rs 107000.00 Total Projected Value: Rs 162649.59

4.3 Description of Database Tables

- users – Stores login credentials and personal details of all registered investors.

- investments – Central table storing all investment records with type, principal, rate, tenure, and calculated maturity amount for each user.
- goals – Tracks user-defined financial goals including target amount, deadline, and achievement status.

CHAPTER 5 – TESTING

Test Case	Input	Expected Output	Result
TC01 – User Registration	Valid name, email, password	Account created successfully	Pass
TC02 – Duplicate Email	Already registered email	Error: "Email already exists"	Pass
TC03 – Login with correct credentials	Valid email & password	Dashboard loads	Pass
TC04 – Login with wrong password	Invalid password	Error: "Invalid credentials"	Pass
TC05 – Add FD Investment	P=50000, R=7, T=3	SI = ₹10,500 / Maturity = ₹60,500	Pass
TC06 – Add Mutual Fund (CI)	P=1,00,000, R=12, T=5	Maturity = ₹1,76,234	Pass
TC07 – Set Financial Goal	Target=5,00,000, Date=3 yrs	Monthly savings suggestion shown	Pass
TC08 – Delete Investment	Select & delete an investment	Record removed from DB and portfolio	Pass
TC09 – View Portfolio Summary	Multiple investments added	Correct total value displayed	Pass
TC10 – Admin View All Users	Admin login	All registered users listed	Pass

CHAPTER 6 – ADVANTAGES AND DISADVANTAGES

6.1 Advantages

- Centralized Management – All investments are tracked in one place, eliminating the need for multiple spreadsheets.
- Automatic Return Calculation – Eliminates manual calculation errors for FD, RD, and compound interest investments.

- Goal Tracking – Helps users plan systematically toward financial targets.
- User-Friendly Interface – Simple Java Swing GUI requires no technical expertise to operate.
- Data Security – User credentials are stored with password hashing.
- Multi-Investment Support – Handles FD, RD, Stocks, Mutual Funds, and Gold in a single system.

6.2 Disadvantages

- No Real-Time Data – Stock and mutual fund prices are entered manually; no live market API integration.
- Desktop-Only – The application is not accessible via web or mobile browsers.
- Single Currency – Currently supports only Indian Rupee (INR); no multi-currency support.
- No Financial Advice – The system only calculates returns; it does not provide personalized investment advice.
- Internet Dependency – Requires a local MySQL server; no cloud database support in the current version.

CHAPTER 7 – APPLICATIONS

The Investment Planner system can be applied in various real-world scenarios:

- **Personal Finance Management** – Individuals can track all their savings instruments and plan for retirement, education, or home purchase.
- **Small Financial Advisory Firms** – Advisors can use the system to maintain client portfolios.
- **Educational Institutions** – Can be used as a practical finance tool to teach students about investment planning and returns.
- **Bank Branch Offices** – Branch staff can use it to quickly calculate and present FD/RD returns to customers.
- **Family Budget Planning** – Households can plan joint investments and monitor shared financial goals.

CHAPTER 8 – CONCLUSION

The Investment Planner is a practical and efficient Java-based application that simplifies the process of managing personal financial investments. The system successfully demonstrates the use of Core Java, Java Swing, JDBC, and MySQL to build a fully functional desktop application. By automating return calculations, enabling goal tracking, and providing a

consolidated portfolio view, the system reduces manual effort and empowers users to make informed financial decisions.

The project fulfills all the defined objectives — from user registration and investment management to report generation and goal tracking. Future enhancements such as live stock market API integration, cloud database support, and a mobile-compatible version would significantly increase the system's reach and usefulness. Overall, this project provides a strong foundation for a real-world financial planning tool.

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